A/B Testing Analysis Using Frequentist and Bayesian Inference

---

1. Executive Overview

This project evaluates whether a newly designed landing page improves user conversion rates relative to the existing version. A randomized controlled A/B test was simulated from 294,478 logged rows, which after cleaning give 290,584 unique users split almost exactly evenly between treatment (new page, 145,310 users) and control (old page, 145,274 users) groups over a 23-day period.

The **primary metric** is the user conversion rate, defined as the proportion of visitors completing a predefined action (e.g., account sign-up). The experiment is designed with a **5% significance level** and **80% statistical power** to detect a **minimum detectable effect (MDE) of 0.75 percentage points**.

This report applies both **Frequentist** and **Bayesian** inferential frameworks. The Frequentist approach relies on classical hypothesis testing using proportions, confidence intervals, and p-values. The Bayesian approach models uncertainty through posterior distributions and evaluates the probability that the treatment outperforms the control under the observed data.

The goal is to derive evidence-based recommendations that guide landing page deployment, optimizing for both **statistical rigor** and **business decision-making** under uncertainty.

---

2. Introduction: Concepts, Context, and Roadmap

This project investigates a central question in conversion rate optimization:
Does a redesigned landing page improve user conversion compared to the existing version?

To explore this, we simulate an experiment using a publicly available dataset from Udacity's A/B testing course. Although the experiment was not conducted firsthand, we reconstruct its context and structure analytically to reflect what a proper real-world experiment would look like.

Report Structure:

This report is structured as follows:

- Theoretical Background
  A concise overview of A/B testing principles, hypothesis testing frameworks, and key statistical concepts.
- Simulated Experiment Reconstruction
  Details of the simulated setup, including randomization, assumptions, success criteria, sample size planning, and validity checks.
- Data Preparation
  Initial exploration, cleaning, and transformation of the dataset to ensure consistency, remove anomalies, and construct relevant variables for analysis.
- Frequentist Analysis
  Application of classical hypothesis tests, confidence intervals, and validation of assumptions.
- Bayesian Analysis
  Probabilistic modelling using Beta priors and posterior inference to quantify uncertainty and support decision-making.
- Expanded Analysis: Regional Performance
  A deeper, country-specific analysis conducted to uncover any regional performance variations.
- Comparative Observations
  Examination of agreements and divergences between Frequentist and Bayesian results, with practical implications.

- Conclusion and Recommendations
  Synthesis of findings and guidance for future testing and deployment.

---

3. Theoretical Background

A/B testing is a controlled experimental methodology used to compare two variants—typically a control and a treatment group—to determine whether a proposed change has a statistically significant effect on a key performance metric. In digital contexts, this typically involves randomly assigning users to different experiences and measuring conversion outcomes to infer causal impact.

Key Concepts:

- Randomization
  Random assignment of users to groups eliminates selection bias and ensures group comparability under the assumption of exchangeability. This enables valid causal inference.
- Hypothesis Testing
  - The **null hypothesis (H_0)** assumes no difference in conversion rates between control and treatment.
  - The **alternative hypothesis (H_1)** posits that a difference exists, either in a one-sided or two-sided form depending on the business question.
- Type I and Type II Errors
  - **A Type I error (α)** occurs when a true null hypothesis is incorrectly rejected **(false positive).** It is typically controlled at **5%**.
  - **A Type II error (β)** occurs when a false null hypothesis fails to be rejected **(false negative).** The **statistical power** of a test, defined as **$1-\beta$**, is often targeted at **80%.**
- Minimum Detectable Effect (MDE)
  The smallest effect size considered practically significant. It is used in conjunction with power and significance level to determine the required sample size for the experiment.
- Frequentist Inference
  - Relies on **test statistics** (e.g., z-test for proportions), **p-values**, and **confidence intervals** to assess evidence against **H_0**.
  - A **p-value** quantifies the probability of observing a result as extreme as the one obtained, under the assumption that **H_0** is true.
  - A **confidence interval** provides a range of plausible values for the true effect size, with a specified confidence level (e.g., 95%) under repeated sampling.
- Bayesian Inference
  - Models uncertainty using **probability distributions** over unknown parameters, updating **prior** beliefs with observed data via **Bayes' theorem**.
  - The **posterior distribution** reflects updated beliefs about the true conversion rates and the treatment effect.
  - **Credible intervals** represent ranges within which the parameter lies with a specified posterior probability (e.g., 95%).
  - Decision-making can be guided by evaluating the **posterior probability** that the treatment outperforms control, yielding more intuitive probabilistic conclusions.

Primary Metric:

- Conversion Rate

The proportion of users in each group who complete a predefined target action (e.g., account sign-up). This is a binary outcome summarized as a proportion, suitable for both proportion-based hypothesis testing and Beta-Binomial modelling in the Bayesian framework.

---

4. Simulated Experiment Reconstruction

Goal of the Experiment:

To evaluate whether a **new landing page** increases the **conversion rate** compared to the existing **old landing page**.

A. Experiment Overview

Feature	Value
Population Size	294,478 logged rows from 290,584 unique users
Groups (after cleaning)	Treatment (new page): 145,310 users Control (old page): 145,274 users
Conversion Metric	Binary converted column (1 if converted, else 0)
Conversion Rate	Control: 12.04% Treatment: 11.88%
Date Range	Jan 2, 2017 → Jan 24, 2017 (23 days)

B. Experiment Setup

Component	Details
Objective	Test if the new landing page leads to significantly different conversion than the old one.
Primary Metric	Conversion rate (binary outcome: converted)
Secondary Metrics	Session duration (if available), bounce rate (not in dataset, could be suggested for future experiments)
Hypotheses	**- Null**: p_new = p_old **- Alternative**: p_new ≠ p_old (two-sided) **- Significance Level:** α = 0.05 **- Power:** 80% **- Minimum Detectable Effect (MDE):** 0.75 percentage points **- Randomization Unit:** user_id **- Randomization Mechanism:** 50-50 random assignment to treatment or control group **- Assignment Logic:** Users assigned to either control (old page) or treatment (new page) group, and saw corresponding landing page

C. Sample Size Calculation Assumptions

- **Baseline Conversion Rate (control):** ~12.04%
- **Target Lift:** +0.75 percentage points (e.g., increase from 12.04% to 12.79%)
- **Significance Level**: 0.05

- **Power:** 0.8

Note: Sample size was not planned prospectively in this dataset. These values reflect what would have been required had the experiment been prospectively designed under the listed assumptions.

Feature	Value
Traffic Allocation	~50% to each group
Unit of Analysis	Individual users (one row per user after cleaning; user-level randomization)
Landing Pages	old_page, new_page

D. Experiment Validity Checks

These checks were run on the cleaned data before interpreting the results:

- **Sample ratio mismatch**: a chi-square test of the group sizes (145,274 vs 145,310) against a 50/50 split gives p = 0.95, so the planned split held.
- **Group/page mismatches**: 3,893 rows (1.3% of the raw data) show a group and landing page that disagree. These rows were removed (see Data Preparation), and a sensitivity check confirms they do not change the result.
- **Novelty effect**: the treatment effect is stable across the experiment weeks (table below), so there is no sign of an effect that fades or grows over time.
- **Randomization check**: beyond group sizes, no user-level covariates (e.g., device, demographics) are available, so balance on other characteristics cannot be validated directly.
- **No significant peeking or early stopping**: Assume one-time evaluation

Week	Control users	Treatment users	Control rate	Treatment rate	Difference
1	42,351	42,573	11.95%	11.78%	−0.16 pp
2	46,306	46,277	11.96%	11.80%	−0.16 pp
3	46,147	46,146	12.14%	12.00%	−0.14 pp
4 (2 days)	10,470	10,314	12.29%	12.13%	−0.16 pp

Note: The mismatched rows were removed at row level, not user level. Every one of the 3,893 affected users also has a matching row, so they remain in the main analysis: 1,998 of them saw both pages and 1,895 saw one page under both group labels. Re-running the test without these users gives a difference of −0.14 pp (p = 0.23, 95% CI [−0.00382, +0.00093]), which leads to the same decision.

E. Realistic Simulated Experiment Narrative

"We simulate a controlled online **A/B test** conducted over 23 days (Jan 2–24, 2017) to evaluate a new landing page design. Users were randomly assigned to either the **treatment group (new page)** or the **control group (old page)**, and their conversions were recorded as **binary** outcomes. The experiment targeted a **minimum detectable effect of 0.75 percentage points**, with **80% power** and a **5% significance level**. In total, **294,478 rows** were logged, and after cleaning **290,584 unique users** form the basis of our analysis."

---

5. Data Preparation

Exploratory Data Analysis (EDA):

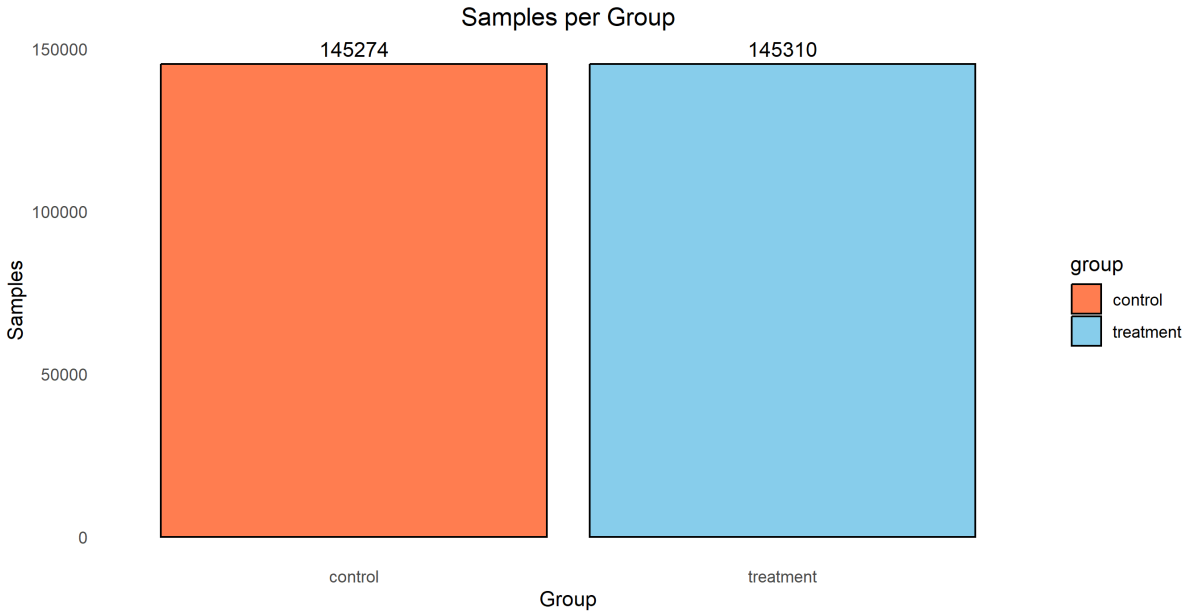
- Data Structure
    - **user_id:** Unique identifier per user.

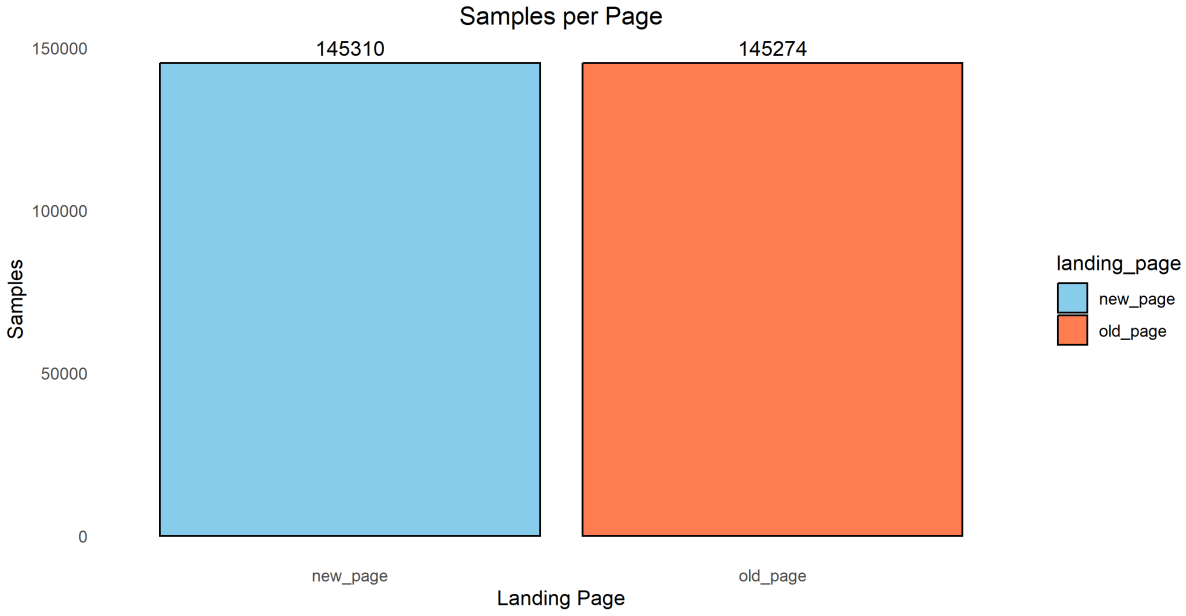
- **timestamp:** Date and time of the user's visit.
- **group:** Assigned experimental condition (control or treatment).
- **landing_page:** Page version seen (old or new).
- **converted:** Binary indicator of conversion (1 = converted, 0 = not converted).

- Data summary
  - The A/B test spanned 23 days (January 2–24, 2017).
  - Before cleaning, the control and treatment groups contain 147,202 and 147,276 rows, respectively.
  - The old and new landing pages appear 147,239 and 147,239 times. These counts differ from the group counts, indicating group-page mismatches.
  - The overall conversion rate across all rows is approximately 11.97%.
- Missing Values
  No missing values were detected during preliminary checks.
- Initial Group and Landing Page Distributions
  Inspection of the group and landing page distributions revealed imbalances due to mismatches between assigned groups and observed landing pages.

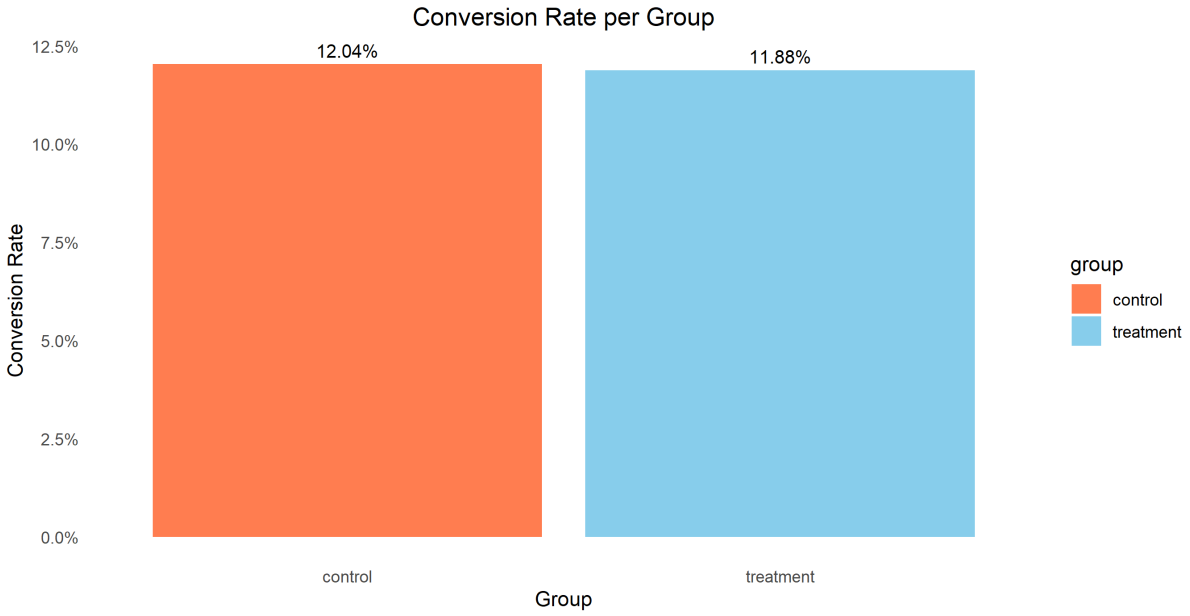
Data Cleaning & Feature Engineering:

- Mismatch Removal
  1,928 control-group rows landed on the new page, and 1,965 treatment-group rows landed on the old page. These 3,893 rows were excluded to preserve the integrity of the experimental design. Each affected user also has a matching row, so the users themselves remain in the analysis; a sensitivity check without them is reported in Section 4D.
- Duplicate Removal
  After mismatch removal, a duplicate entry remained for user_id 773192. The earliest exposure was kept, leaving one row per user: 290,584 users in total.
- Variance Stability Across Groups
  The outcome variance is consistent between the treatment and control groups, so the pooled z-test is appropriate. For a binary outcome the variance follows from the conversion rate, so this is not separate evidence about randomization; the sample ratio check in Section 4D covers the group split.
- Post-cleaning Distributions
  User allocation between groups remained stable and balanced after mismatch removal (145,274 control, 145,310 treatment). Landing page distributions also reflected correct assignment post-cleaning.



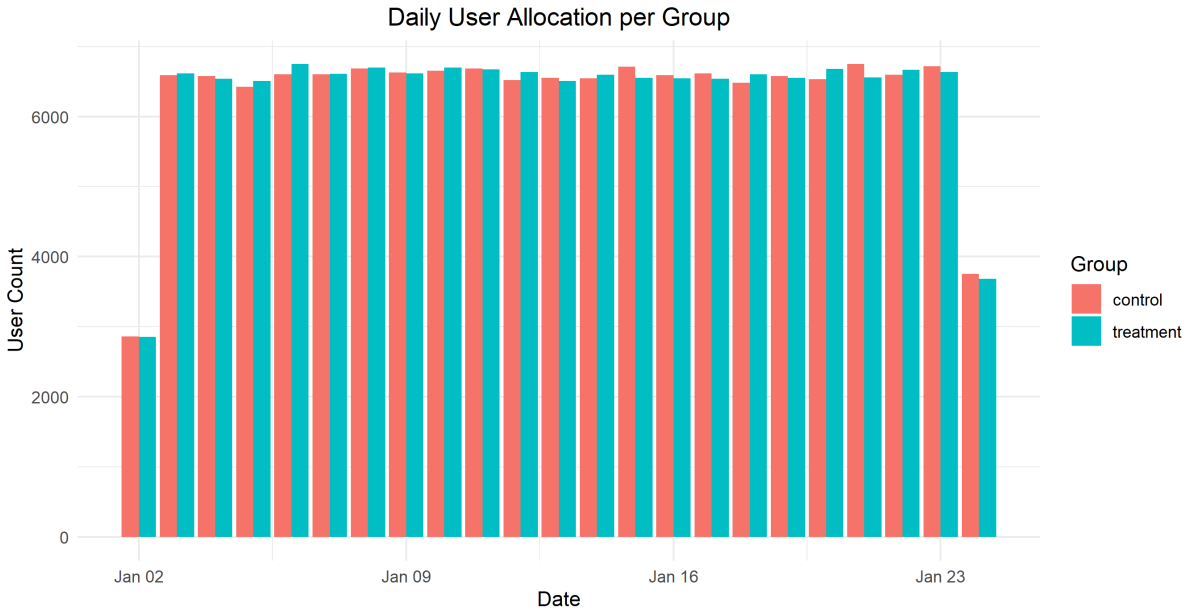


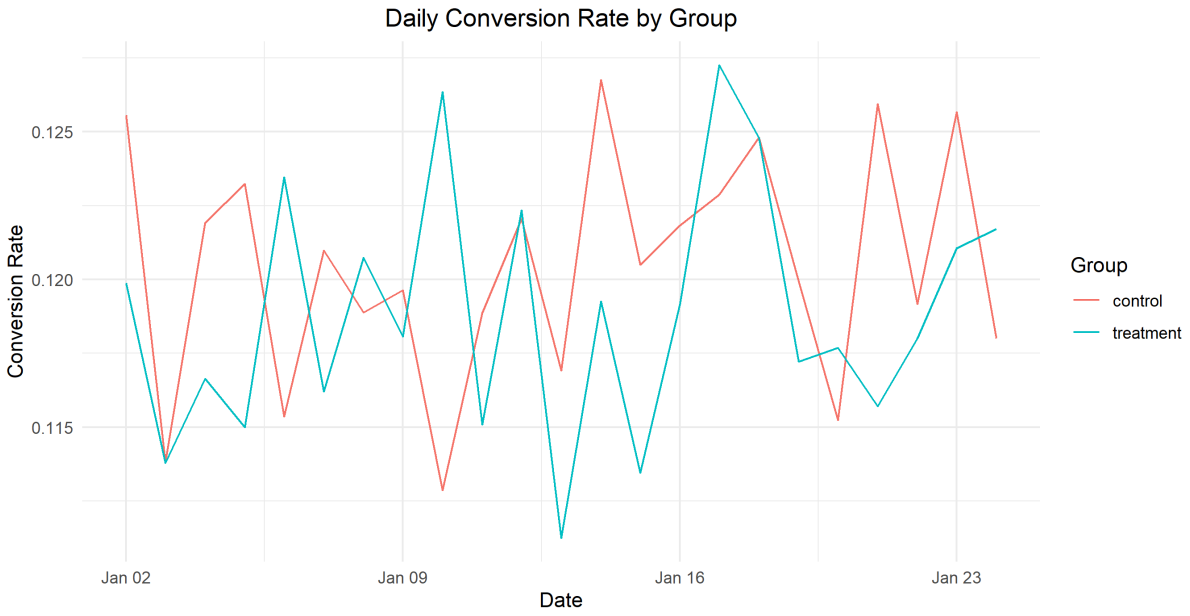
- Conversion Rate Analysis
  The control group had a conversion rate of 12.04%, while the treatment group had a rate of 11.88%. Conversion rates by landing page match those by group, confirming consistency after cleaning.



- Dataset Limitations
  Due to dataset constraints, no additional user-level covariates (e.g., demographics, device type) are available to assess the randomization balance beyond group sizes. This limits the ability to fully rule out potential confounding, though data integrity checks (duplicate removal, sample ratio, timestamp consistency) support the validity of the analysis.

- Temporal Trends
  Daily group allocation remained consistent across the experiment. Conversion rates fluctuated over time: the control group peaked on January 14 and bottomed on January 10. The treatment group peaked on January 17 and was lowest on January 13. These fluctuations appear random and do not indicate systematic bias.





---

6. Frequentist Analysis

Define the Problem and Formulate Hypotheses:

- Research Question
  Test if the new landing page leads to significantly different conversion than the old one.
- Stating Hypotheses
  - **Null hypothesis (H_0):** The new landing page leads to no different conversion than the old one.
  - **Alternative hypothesis (H_1):** The new landing page leads to significantly different conversion than the old one.

Power Analysis Setup:

- **Baseline conversion rate (p_1):** 12.04% (0.1204)

- **Target conversion rate ($p_2 = p_1 + MDE$):** 12.79% (0.1279)
- **Minimum Detectable Effect (MDE):** 0.75 percentage points absolute uplift planned (from 12.04% to 12.79%)
- **Significance level (α):** 0.05 (two-sided)
- **Power:** 80%
- **Required sample size per group:** ≈ 30,342
- **Actual sample sizes:** 145,274 (control), 145,310 (treatment), well above the minimum

Observed results:

- **Observed conversion rates**: 12.04% (control), 11.88% (treatment).
- **Observed difference**: treatment conversion is 0.16 percentage points lower than control (−0.0016).
- **Pooled proportion:** pp ≈ 0.1196
- **Standard error:** SE ≈ 0.0012
- **Test statistic:** z ≈ −1.31
- **Corresponding two-sided p-value:** ≈ 0.19
- **95% Confidence Interval:** [−0.00394, +0.00078]

Interpretation:

- Despite large sample sizes and sufficient statistical power to detect the pre-specified minimum detectable effect (MDE), the observed uplift in conversion was **negative** (−0.16 percentage points) and **not statistically significant**.
- The **95% confidence interval** for the difference in conversion rates **(−0.39 to +0.08 percentage points)** includes **zero**, indicating that the observed effect may be due to chance.
- The upper end of the interval (+0.08 percentage points) lies far below the planned MDE of +0.75 percentage points. A lift of the planned size is therefore **ruled out**, not merely undetected.
- The **p-value exceeds the 5% significance threshold**, providing **no statistical evidence** to reject the null hypothesis of equal conversion rates between control and treatment groups.
- **The treatment does not lead to a measurable improvement in conversion**. If anything, it may have a slight detrimental effect — but this effect is both **small in magnitude** and **statistically indistinguishable from random variation**.

---

7. Bayesian Analysis

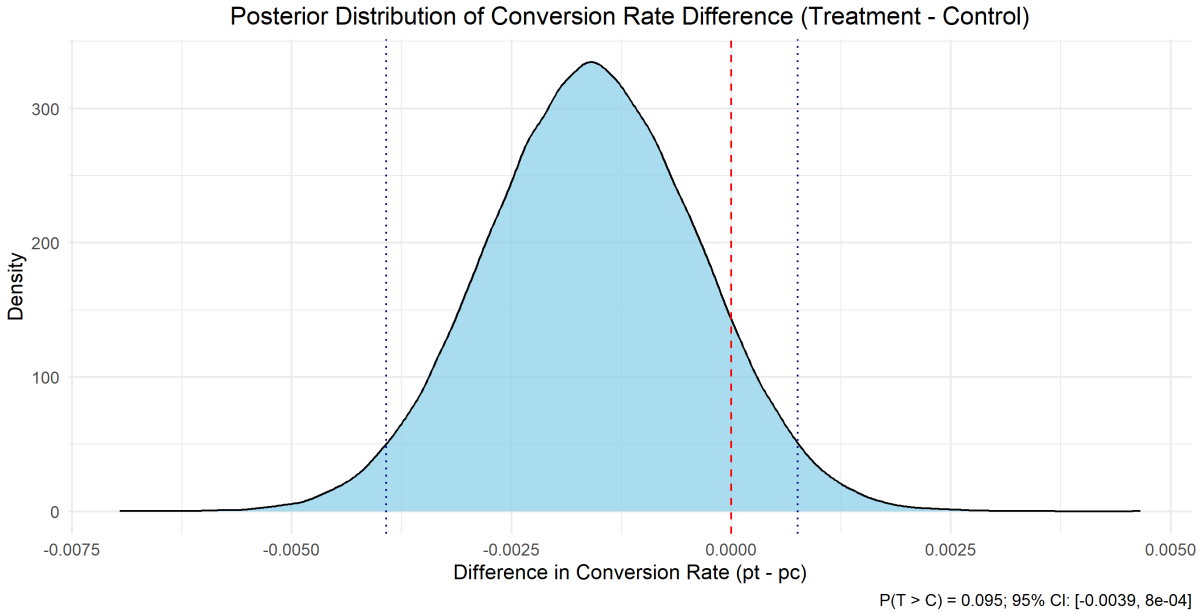
- Set prior distributions
  We assume uninformative priors for both group conversion rates:
  - $pc \sim Beta(1,1)$
  - $pt \sim Beta(1,1)$

Note: We use Beta(1,1) as a flat, uninformative prior, reflecting no prior knowledge and letting the data dominate. In smaller samples, a weakly informative prior can improve stability and guard against implausible estimates. Here, large sample sizes justify a noncommittal choice.

- Posterior distributions
  Upon observing the data:

- pc|data ~ Beta(17490, 127786)
- pt|data ~ Beta(17265, 128047)

- Sample from posteriors
  To approximate the posterior distribution of the difference in conversion rates, we draw (with a fixed random seed, so the results are reproducible):
  - 100,000 samples from Beta(17490, 127786) for the control group.
  - 100,000 samples from Beta(17265, 128047) for the treatment group.
- **Compute the elementwise difference:** This provides a Monte Carlo approximation of the posterior distribution of the **treatment effect**, δ = pt − pc. The posterior distribution of the difference in conversion rates is shown below.



- **Summary statistics**
  - Posterior probability that treatment outperforms control: P(pt > pc) ≈ 9.5%
  - 95% credible interval for pt − pc: [−0.0039, +0.0008]
  - Posterior probability that the lift reaches the planned MDE (+0.75 percentage points): ≈ 0%
- **Region of Practical Equivalence (ROPE):**
  We define a **ROPE** around zero to represent effect sizes considered practically negligible
  - **ROPE:** [−0.003, +0.003]
  - Posterior mass within **ROPE**: ~88.2%

Note: The ROPE threshold was selected based on practical significance—i.e., changes smaller than ±0.3 percentage points are not actionable given cost, risk, and impact. This allows us to assess not just *whether* there's a difference, but whether it's **meaningful**.

Interpretation:

- Only ~9.5% of the posterior distribution lies above zero, providing **weak evidence** in favour of the treatment.
- The **95% credible interval [−0.0039, +0.0008]** is **centred slightly below zero** and includes both practically negative and practically null effects. This implies that the treatment may be

slightly worse than control or have no real effect.

- With ~88.2% of the posterior mass within the ROPE and essentially no chance of reaching the planned MDE, we have **moderate evidence** that the new page's effect is too small to be meaningful, providing a quantitative basis for the decision not to deploy it.

---

8. Expanded Analysis: Regional Performance

While the overall A/B test showed no significant difference, a deeper, country-specific analysis was conducted to uncover any regional performance variations. Country data were joined to the cleaned users with no users lost. Each country was analysed with a Bayesian posterior comparison and a Frequentist z-test, with a Holm correction for testing 3 countries. The Canadian segment is the smallest (14,499 users), so its estimates are the least precise.

Our analysis of the new page's conversion rate in each country revealed the following:

Country	Users	Difference	P(new better)	95% credible interval	ROPE	p-value	Holm p
US	203,619	−0.22 pp	6.6%	[−0.00496, +0.00064]	72.1%	0.13	0.40
UK	72,466	+0.11 pp	68.0%	[−0.00362, +0.00590]	73.6%	0.63	0.63
CA	14,499	−0.69 pp	9.9%	[−0.01728, +0.00361]	20.2%	0.19	0.40

No country shows a statistically significant effect, even before the correction. A logistic regression test of whether the treatment effect differs by country (treatment × country interaction) gives p = 0.29, so there is no evidence that the new page performs differently in any region.

The UK's 68.0% probability of improvement should not be read as evidence of a lift: its credible interval is wide and centred close to zero, and its z-test p-value is 0.63. With 3 noisy country estimates, one of them leaning positive is not surprising by chance alone. The low ROPE coverage in Canada is a result of the smaller sample size, which leads to a wider credible interval and higher uncertainty, not a sign of a meaningful effect.

---

9. Comparative Observations

Post-Hoc Observations and Power (Frequentist Analysis):

Frequentist hypothesis testing revealed no statistically significant difference between the treatment and control groups. The observed −0.16 percentage point difference in conversion rates is far from the pre-specified minimum detectable effect (MDE) of +0.75 percentage points, and the associated p-value of 0.19 exceeds the conventional 5% significance threshold.

The 95% confidence interval for the difference in conversion rates [−0.00394, +0.00078] includes zero and skews negative, reinforcing the lack of statistical evidence for a meaningful treatment effect. The experiment was not underpowered: it had roughly 5 times the sample needed to detect the planned MDE, and the interval excludes a lift of that size.

From a decision-making perspective, these results provide no justification to implement the treatment. The evidence supports a conservative stance: the new page is unlikely to outperform the current version, and it may even slightly underperform. In light of these findings, the prudent course is to halt rollout and re-evaluate the proposed changes.

Post-Hoc Observations and Decision Limits (Bayesian Analysis):

Posterior inference yielded a 9.5% probability that treatment outperforms control, and a 95% credible interval of [−0.0039, +0.0008], which includes zero and skews negative.

Using a ROPE of ±0.3 percentage points, approximately 88.2% of the posterior mass falls within this region. This suggests that the treatment effect is likely inconsequential from a business perspective.

In Bayesian terms, this does not provide evidence in favour of the treatment. Rather, it provides moderate evidence for practical equivalence — i.e., that the new page is no better (and possibly worse) than the old.

Although Bayesian methods avoid binary hypothesis testing, this result supports the same operational decision: there is insufficient evidence to justify deploying the new page. The minor −0.16 percentage point observed drop, combined with high ROPE coverage and low probability of improvement, points toward conservatism and re-evaluation rather than rollout.

---

10. Conclusion and Recommendations

Decision:
Do not deploy the new landing page at this time.

Reasoning:

- Both **Frequentist** (p-value = 0.19, CI includes 0) and **Bayesian** (P(pt > pc) ≈ 9.5%) results suggest that the new page **does not improve conversion** — and may even underperform.
- A large share of the **posterior distribution falls within the ROPE** (Region of Practical Equivalence), indicating a high probability that the difference is practically negligible.
- The experiment had ample power: the confidence interval tops out at +0.08 percentage points, so **a lift of the planned 0.75 percentage points is ruled out**.
- The result holds under the validity checks: the 50/50 split held, the effect is stable week to week, and removing users with mismatched rows gives the same decision.
- The expanded regional analysis supports this conclusion. No country shows a significant effect, and there is no evidence that the effect differs by country (interaction p = 0.29). The UK's 68.0% probability of improvement is consistent with chance.

Next Steps:

- **Keep the current page** in all markets, including the UK.
- **Redesign and retest**: Consider changes to layout, content, or CTA strategy, and run a new A/B test if the new design justifies it, with a pre-registered MDE and a fixed horizon.
- **Broaden your metric scope**: Track revenue per user or lifetime value alongside conversion, and secondary metrics (e.g., session duration, click-throughs, bounce rate) to better understand user engagement beyond binary conversion.
- **Cost-benefit assessment**: Evaluate whether pursuing further design iterations is worth the **engineering cost, opportunity cost, and potential lift**.